VEHICLE SERVICE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

RAMANATHAN S

(241401076)

in partial fulfilment of the End Semester Examination for the course

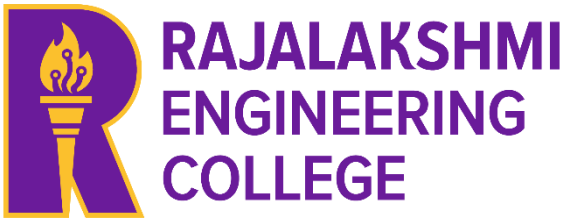
CB23PE1 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS (CSBS)



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “Vehicle Service Management System” is the bonafide work of RAMANATHAN S (241401076), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23PE1 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, RAMANATHAN S (241401076), hereby declare that the project report titled "Vehicle Service Management System", submitted for the End Semester Examination of the course CB23PE1 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 01 November 2026

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

RAMANATHAN S

ABSTRACT

Vehicle service centres in small and mid-sized towns still record customers, vehicles and service jobs in registers, job cards and phone messages. Customer and vehicle details are written again for every visit, bookings are tracked on paper, mechanics are assigned verbally, and the owner has no live view of how many vehicles are booked, which jobs are finished or how much revenue has been earned. These gaps lead to missed appointments, billing errors and no clear accountability for each job.

This project presents the Vehicle Service Management System, a low-code application built on Zoho Creator to replace that manual process. The application contains four forms: customer details, Vehicle Details, Service Booking and Service Records. The forms are linked by lookup fields, so a vehicle belongs to a customer, a booking belongs to a customer and a vehicle, and a service record belongs to a booking. Four reports (All Customer Details, All Vehicle Details, Service Booking Report and All Service Records) present the data. The Vehicle Service Dashboard page shows KPI tiles for the customer count, vehicle count, service requests and total sales revenue, a bar chart of vehicles by type, a bar chart of bookings by month, a pie chart of service types and a donut chart of jobs by mechanic. A form workflow calculates the delivery date from the service package, sets the booking status to Booked when it is left blank and sends a booking confirmation e-mail to the vehicle owner with a Deluge sendmail task. Access is controlled through two users, an Administrator and a Mechanic.

Thirty-three customers, 33 vehicles, 32 service bookings and 32 service records populate the application, and the dashboard shows a total sales revenue of ₹519,700.00. Testing covered form submission, the lookup relationships, the report record counts, the dashboard figures and the two user accounts. The application was created on the Zoho Creator trial plan on 01 November 2026 and is available at https://creatorapp.zoho.in/ramanathan.s.2024.csbs_ra271/vehicle-service-management-system/. The result is a single, shared and always current view of the service centre in which paper registers and manual totals are replaced by linked forms, one confirmation e-mail and one dashboard.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Dashboard Analytics; Vehicle Service Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Vehicle Service Management System
Application Domain	Admin / Mechanic
Student Name & Register No.	Ramanathan S – 241401076
Zoho Creator Edition	Trial
Application Name (Link Name)	Vehicle Service Management System
Published / Live URL	https://creatorapp.zoho.in/ramanathan.s.2024.csbs_ra271/vehicle-service-management-system/#Page:Vehicle_Service_Dashboard
Demo Credentials (per role)	Admin: ramanathan.s.2024.csbs@rajalakshmi.edu.in | Mechanic: ramanathan012006@gmail.com
Number of Forms / Subforms	4 / 0 (customer details, Vehicle Details, Service Booking, Service Records)
Number of Reports (by type)	List: 4 Kanban: 0 Calendar: 0 Pivot/Chart: 0 Others: 0
Number of Pages / Dashboards	1 (Vehicle Service Dashboard)
Number of Workflows	Form: 1 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	No standalone custom function. The form workflow runs one Deluge script on booking submission: it looks up the service package, works out the delivery date, sets the status to Booked and sends a confirmation e-mail.
Integrations	Zoho Creator mail service (Send Mail action) for the booking confirmation e-mail
Roles & Profiles Defined	Administrator (application owner) and Mechanic (permission: Mechanic). Two users in total.
Environments Used	Development workspace (published directly); no Stage or Production environment configured
Date of Production Deployment	01 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
INR	Indian Rupee (₹)
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
OTP	One-Time Password
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
VSM	Vehicle Service Management

CHAPTER 1
INTRODUCTION

1.1 Background

Vehicle servicing is the routine of inspecting, repairing and maintaining cars, bikes and scooters so that they stay safe and roadworthy. A service centre depends on accurate customer, vehicle and job records because appointments, cost estimates, mechanic allocation and billing all start from them. This project addresses that process for a realistic multi-vehicle service centre in Chennai, with an Administrator who manages the centre and its records and Mechanics who carry out the jobs.

In the situation studied, owners of cars, bikes and scooters book services such as general service, engine repair, oil change, brake repair, tyre replacement and washing. The front desk writes the owner and vehicle details in a register, notes the problem on a job card, assigns a mechanic verbally and totals the parts and labour cost by hand after the work. The owner is told the amount by phone, and the centre counts its workload and revenue by going through the registers. The Vehicle Service Management System was built on Zoho Creator to move this entire process into one cloud application that every user can reach through a single URL.

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1.2 Problem Statement

Customer and vehicle details are written repeatedly in registers, bookings are not visible in one place, and the owner cannot tell at a glance how many vehicles are booked, which mechanic is handling which job or how much revenue has been earned. As a result appointments are missed, estimates and final bills do not match, and nobody can easily trace a job back to a vehicle or a customer. The service centre needs a central, role-controlled application that captures customers, vehicles, bookings and service work in a standard format, confirms bookings automatically and reports the figures without manual counting.

- Data duplication and inconsistency: the same customer and vehicle are written again for every visit, with different spellings and vehicle number formats.
- Delayed visibility: the owner learns about bookings, workload and revenue only after the registers are collected and totalled.

- Manual reporting: counting vehicles by type, bookings by month and revenue takes repeated filtering and summing.
- No confirmation: customers receive nothing in writing about the service package, cost and expected delivery date.
- No access control: anyone holding the register can edit or remove entries, and there is no record of which mechanic carried out a job.

1.3 Objectives

- To design a linked data model and forms for capturing customers, vehicles, service bookings and service records (customer details, Vehicle Details, Service Booking and Service Records forms, Section 5.2).
- To automate booking confirmation using a form workflow and Deluge scripting (Sections 5.5 and 5.6).
- To provide reports and a dashboard for the administrator and mechanics (four list reports and the Vehicle Service Dashboard, Sections 5.3 and 5.4).
- To send e-mail through Zoho's mail service and to restrict access through the Administrator and Mechanic users (Sections 5.7 and 5.8).
- To publish the application on Zoho Creator with basic governance controls (Chapter 7).

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1.4 Scope of the Project

In scope: capture of customers, vehicles, service bookings and service records through four linked forms; storage of 33 customers, 33 vehicles, 32 service bookings and 32 service records; four list reports (All Customer Details, All Vehicle Details, Service Booking Report and All Service Records); the Vehicle Service Dashboard page with KPI tiles and charts; a booking confirmation e-mail workflow; and access for an Administrator and a Mechanic. The platform is the Zoho Creator web application.

Out of scope: a customer portal, online payment collection, approval processes, Blueprints, scheduled reminders, SMS notifications, spare-part inventory and AI features such as Zia. The Zoho Creator mobile app is supported by the platform but was not configured or tested for this project; the Portal and Mobile panels of the application summary are empty (Figure 5.1).

1.5 Justification for a Low-Code Approach

A vehicle service management system is a data-centric, form-and-report application with a small set of business rules, which is exactly the class of problem for which low-code platforms are intended [1], [2]. Zoho Creator provides the form builder, report types, page builder, workflow engine, user management and cloud hosting as built-in services, so the first working version could be delivered in days instead of the weeks that a custom web application would need. Bock and Frank describe this shift as moving development closer to the business users who understand the process, which is also why a service centre can maintain such an application without a dedicated programming team [3].

Low-code was also chosen for cost and risk reasons. There is no server, database or authentication module to install and secure; the application is available on the web immediately after publishing; and Deluge scripting is available for the few places where configuration alone is not enough, such as the delivery-date calculation and the confirmation e-mail. The trade-offs of this choice are discussed in Section 8.4.

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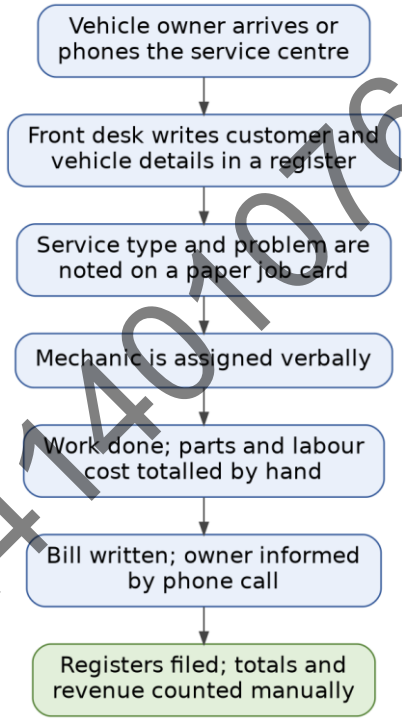
1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

At present the service process is manual and register based. An owner arrives or phones the centre, and the front desk writes the owner and vehicle details in a register and the problem on a paper job card. A mechanic is assigned verbally, and after the work the parts and labour cost are added up by hand and the amount is told to the owner by phone. At the end of the day or month the registers are read again to count vehicles, jobs and revenue. Figure 2.1 shows this flow.



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Figure 2.1: As-Is Process Flow of the Existing Vehicle Service Process

2.2 Limitations of the Existing System

- The same customer and vehicle are written many times in different registers, so the data is duplicated and often inconsistent.
- The owner sees bookings and workload only after the registers are collected, which delays planning.
- Totals for vehicles, service requests and revenue must be calculated manually every time.

- There is no automatic confirmation or audit trail; a job cannot be traced to a mechanic with certainty.
- Access to the data cannot be restricted by role, and the registers are available only where they are kept.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	8–10 weeks for design, coding, database, login and hosting setup (estimate)	About 1–2 days for the forms, reports, dashboard and workflow (estimate)
Skills required	Full-stack developers (front end, back end, database, authentication)	Creator builder skills and basic Deluge scripting
Hosting & infrastructure	Own server or cloud account, database, SSL and domain	Hosted by Zoho at creatorapp.zoho.in; no infrastructure to manage
Mobile application	Separate responsive site or native app to be built	Platform provides the web app and Zoho Creator mobile app from the same definition
Security & access control	Login, roles and sharing rules to be coded and tested	Built-in users, permissions and sharing options
Maintenance & change	Code changes, testing and redeployment for every change	Changes made in the browser builder and applied at once
Customisation limits	Unlimited, any design or logic can be coded	Limited to builder options and Deluge limits; code needed only for special logic
Cost	Developer salaries, hosting and maintenance costs	Trial plan during development; per-user subscription afterwards

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2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide	Explains low-code concepts, citizen development and platform choices	Course reference for the approach and terminology used
2	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, users and APIs	Source for configuring every component
3	A. C. Bock and U. Frank, 2021 [3]	Low-Code Platform (BISE)	Defines low-code platforms and discusses citizen developers and productivity	Justifies the low-code approach and maintainability
4	Zoho Corporation [4]	Deluge Script Reference Guide	Describes the syntax of fetch, conditions, date functions and sendmail	Basis for the booking script and the Send Mail action
5	Gartner, Inc., 2024 [5]	Magic Quadrant for Enterprise Low-Code Application Platforms	Evaluates vendors and market trends in low-code platforms	Shows the maturity of the platform class used

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2.5 Emerging Trends in Low-Code Development

Three trends in low-code development influenced this project. The first is citizen development: platforms such as Zoho Creator allow business users to build and change their own applications, which supports the idea that a service centre can maintain the system itself [3], [6]. The second is hyper-automation, the use of workflows and scripts to remove repeated manual work; in this project it appears as the booking workflow that fills in the delivery date and status and sends the confirmation e-mail. The third is AI-assisted development and analytics, including Zoho's Zia assistant. AI features were not needed for the current scope, but the structured service history collected by the application is the foundation for future predictive service reminders (Section 9.2).

2.6 Proposed System

The proposed system is a Zoho Creator application named Vehicle Service Management System. Every limitation listed in Section 2.2 is answered by a specific feature. Duplicated and inconsistent data is replaced by four linked forms with lookup

fields and controlled drop-down values, so a customer and a vehicle are entered once and then selected. Missing confirmation is solved by the booking workflow, which fills in the delivery date and sends an e-mail as soon as a booking is saved. Manual counting is replaced by the four reports and the Vehicle Service Dashboard. Traceability comes from the Service Records form, which stores the mechanic, work, costs and status of every job, and access is limited to the two configured users.

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CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner (ramanathan.s.2024.csbs@rajalakshmi.edu.in) with full control of the application.	Manage users and permissions; configure forms, reports, the workflow and the dashboard; add customers, vehicles and bookings; view all data.
Mechanic	Service technician (ramanathan012006@gmail.com); permission: Mechanic.	Review bookings and vehicles, enter service records with work description, parts cost and labour cost, update service and payment status, and view the reports and dashboard.

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3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a customer to be added with Customer name, Phone (with country code), Email and Address.	Administrator	High
FR-02	The system shall register a vehicle against a customer with Vehicle Number, Brand, Model, Vehicle Type, Fuel Type and Manufacturing Year, using drop-downs for Brand, Vehicle Type and Fuel Type.	Administrator	High
FR-03	The system shall book a service by selecting the customer and vehicle, Service Type, Booking Date, Preferred Time, Problem Description, Estimated Cost and Booking Status.	Administrator, Mechanic	High
FR-04	The system shall record the service work against a booking: Assigned Mechanic, Service Date, Work Description, Parts Cost, Labour Cost, Total Amount, Service Status and Payment Status.	Mechanic, Administrator	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall send a booking confirmation e-mail to the vehicle owner showing the vehicle number, service package, cost, service date and expected delivery date, and shall fill in the delivery date and the status Booked when they are left blank.	Administrator	Medium
FR-06	The system shall list all records in four reports: All Customer Details, All Vehicle Details, Service Booking Report and All Service Records.	All roles	High
FR-07	The system shall display a dashboard with the customer count, vehicle count, service request count, total sales revenue, a vehicle-type bar chart, a bookings-by-month bar chart, a service-type pie chart and a mechanic donut chart.	Administrator	Medium
FR-08	The system shall restrict access to the two configured users, the Administrator and the Mechanic.	Administrator	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to 33 records shall load within 3 seconds on a normal broadband connection.
NFR-02	Security	Users shall access the application only through their Zoho account sign-in and their assigned permission.
NFR-03	Usability	The application shall be usable on the web and, through the platform, on the Zoho Creator mobile app.
NFR-04	Availability	The application shall be hosted on Zoho's cloud and be reachable at its published URL at any time, subject to Zoho service availability.
NFR-05	Maintainability	Forms, reports, the workflow and the dashboard shall be editable in the Creator builder, with meaningful names so that a citizen developer can maintain them.
NFR-06	Auditability	Record changes shall be traceable through the audit trail.

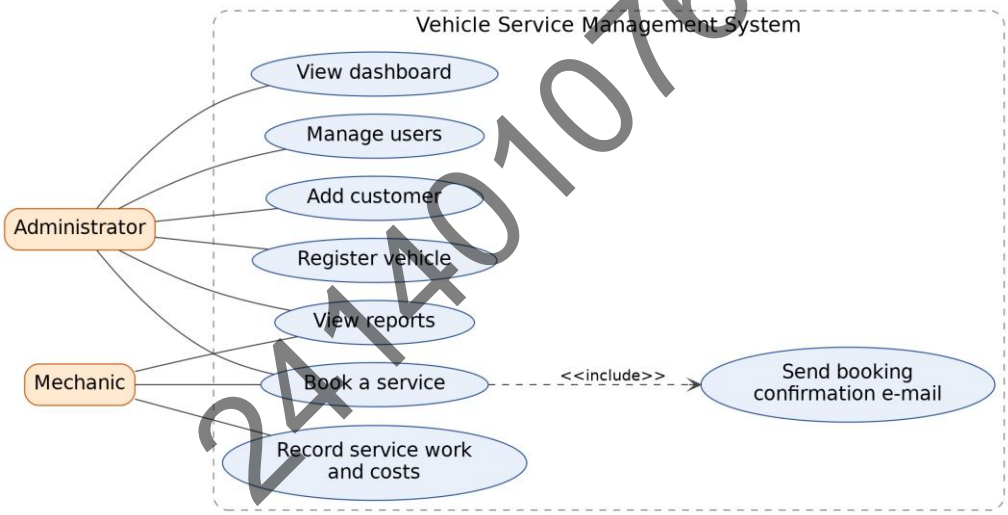
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3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Trial Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	Zoho Creator mail service (Send Mail action)
Mobile	Zoho Creator mobile app (Android / iOS) supported by the platform; not used in this project
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model



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Figure 3.1: Use Case Diagram of the Vehicle Service Management System

Table 3.5: Use Case Description – Book a Vehicle Service

Use Case ID	UC-01
Use Case Name	Book a Vehicle Service
Actor(s)	Administrator (also Mechanic)
Pre-condition	The user is signed in to the Vehicle Service Management System, and the customer and the vehicle have already been added.
Main Flow	1. The user opens Service Booking from the menu. 2. The user selects the customer and the vehicle. 3. The user selects the Service Type. 4. The user enters the Booking Date and selects the Preferred Time. 5. The user enters the Problem Description, the Estimated Cost and the Booking Status. 6. The user clicks Submit. 7. The system saves the record and runs the booking workflow, which fills the delivery date and sends the confirmation e-mail.
Alternate Flow	If the customer or the vehicle does not exist, the user adds it first from the customer details or Vehicle Details form. If a value is not valid for a field, the form shows an error and the record is not created.
Post-condition	The booking appears in the Service Booking Report and in the dashboard counts, and the confirmation e-mail is sent.
Related FR	FR-03, FR-05, FR-06

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CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

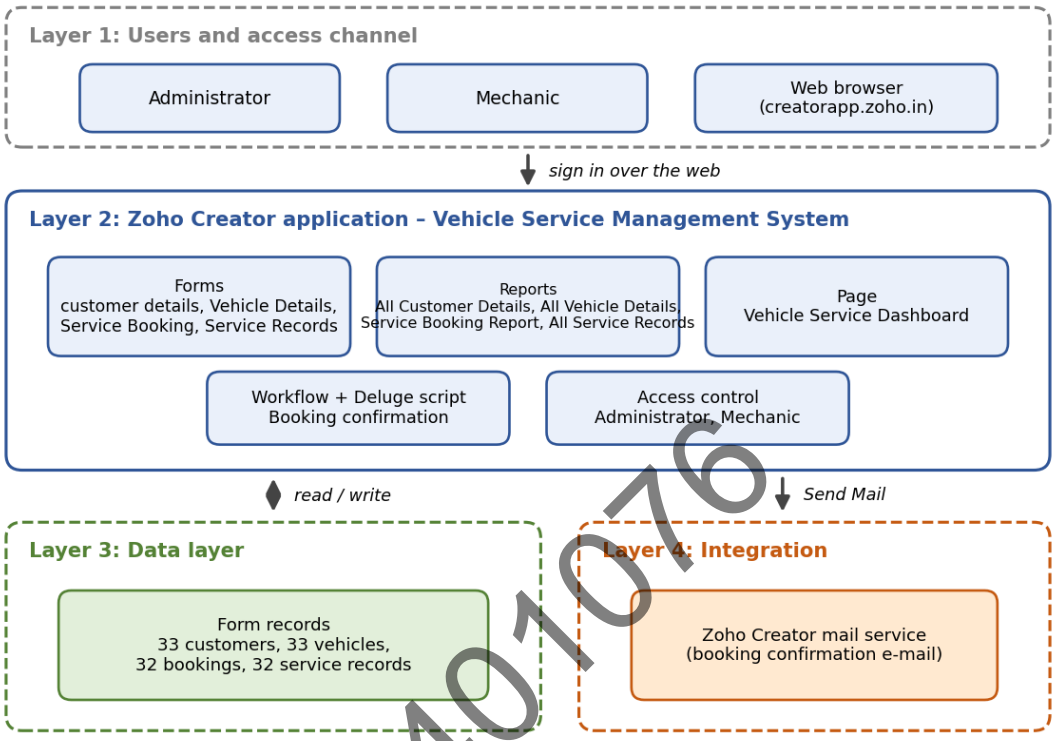


Figure 4.1: System Architecture of the Vehicle Service Management System

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The architecture has four layers. At the top are the users and the access channel: the Administrator and the Mechanic sign in through a web browser to the published URL. The second layer is the Zoho Creator application, which holds the four forms, the four list reports, the Vehicle Service Dashboard page, the booking workflow with its Deluge script and the user access settings. The third layer is the data layer, where every submission is stored as a record of its form and is read by the reports and the dashboard. The fourth layer is the integration: Zoho Creator's mail service sends the booking confirmation e-mail.

4.2 Data Model

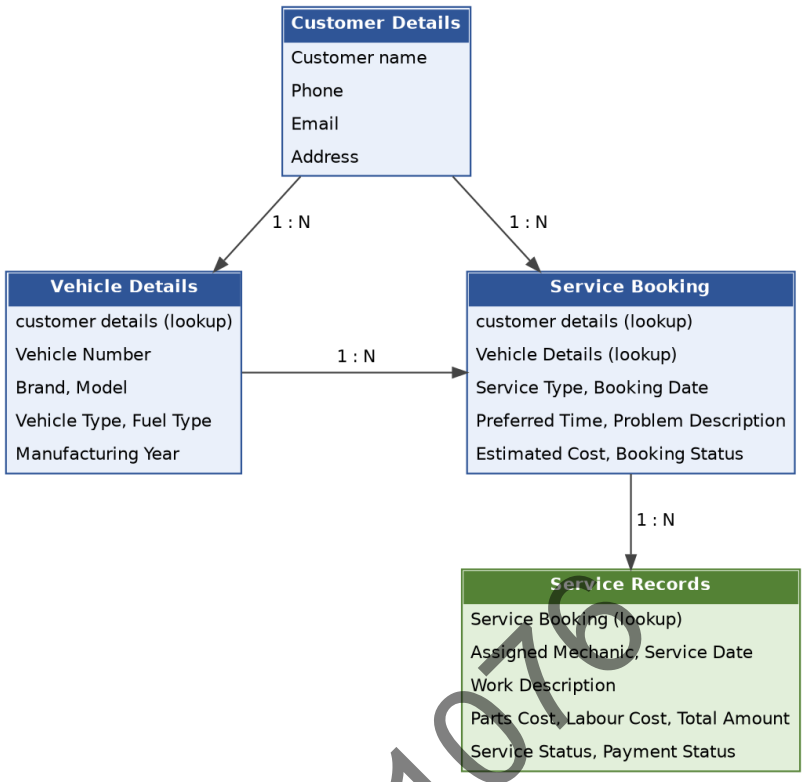


Figure 4.2: Entity–Relationship Diagram of the Vehicle Service Management System

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
customer details	Customer name, Phone, Email, Address	Vehicle Details, Service Booking	One-to-Many	Referenced by lookup fields
Vehicle Details	Vehicle Number, Brand, Model, Vehicle Type, Fuel Type, Manufacturing Year	customer details; Service Booking	Many-to-One (customer); One-to-Many (bookings)	Lookup field customer details
Service Booking	Service Type, Booking Date, Preferred Time, Estimated Cost, Booking Status	customer details, Vehicle Details, Service Records	Many-to-One (customer, vehicle); One-to-Many (records)	Lookup fields customer details and Vehicle Details
Service Records	Assigned Mechanic, Service Date, Parts Cost, Labour Cost, Total Amount, Service Status, Payment Status	Service Booking	Many-to-One	Lookup field Service Booking

The system uses a four-form model linked by lookup fields. A customer is entered once, a vehicle is linked to its owner, a booking is linked to both the customer and the vehicle, and a service record is linked to its booking. No subform was required. All reports and

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the dashboard read these four forms, which keeps one source of truth for every figure shown.

4.3 Form Design

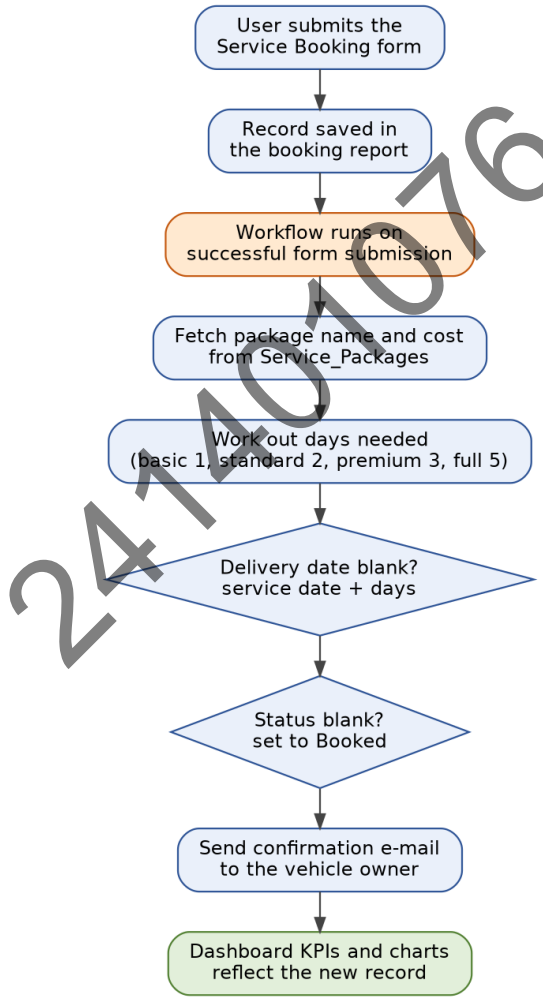
Table 4.2: Field Specification – Four Application Forms

Form	Field Label	Field Type	Remarks
customer details	Customer name	Single Line	Name of the vehicle owner
customer details	Phone	Phone	Country code selector, default +91 (India)
customer details	Email	Email	Used for the booking confirmation e-mail
customer details	Address	Multi Line	Area and city, for example Madipakkam Chennai
Vehicle Details	customer details	Lookup	Owner of the vehicle
Vehicle Details	Vehicle Number	Single Line	Registration number, for example TN39EF2030
Vehicle Details	Brand	Drop Down	Values include BMW, Tata, Hyundai, Kia, Toyota, Honda, Yamaha
Vehicle Details	Model	Single Line	Example: Tiago, Brezza, Splendor Plus
Vehicle Details	Vehicle Type	Drop Down	Car, Bike, Scooter; used by the dashboard bar chart
Vehicle Details	Fuel Type	Drop Down	Petrol, Diesel, Electric, CNG
Vehicle Details	Manufacturing Year	Number	Four-digit year
Service Booking	customer details	Lookup	Customer who books the service
Service Booking	Vehicle Details	Lookup	Vehicle to be serviced
Service Booking	Service Type	Drop Down	General Service, Engine Repair, Oil Change, Brake Repair, Tyre Replacement, Washing
Service Booking	Booking Date	Date	Format dd-MMM-yyyy
Service Booking	Preferred Time	Drop Down	9 AM – 11 AM, 11 AM – 1 PM, 2 PM – 4 PM, 4 PM – 6 PM
Service Booking	Problem Description	Multi Line	Complaint noted at the front desk
Service Booking	Estimated Cost	Currency	Indian rupee format
Service Booking	Booking Status	Drop Down	Status of the booking (Booked when left blank by the workflow)
Service Records	Service Booking	Lookup	Booking to which the work belongs
Service Records	Assigned Mechanic	Single Line	Mechanic who did the work; used by the donut chart
Service Records	Service Date	Date	Format dd-MMM-yyyy

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Form	Field Label	Field Type	Remarks
Service Records	Work Description	Multi Line	Work carried out
Service Records	Parts Cost	Currency	Cost of parts
Service Records	Labour Cost	Currency	Cost of labour
Service Records	Total Amount	Currency	Parts Cost plus Labour Cost; basis of the revenue tile
Service Records	Service Status	Drop Down	Progress of the job
Service Records	Payment Status	Drop Down	Payment state of the job

4.4 Process and Workflow Design



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Figure 4.3: Workflow / Process Flow for Service Booking and Confirmation

Figure 4.3 shows the one automated process of the application. When a user submits the Service Booking form, the record is saved in the booking report and the form workflow runs on successful form submission. The script fetches the package name and cost, works out the number of days needed from the package name, calculates the

delivery date when it was left blank, sets the status to Booked when it was left blank and sends the confirmation e-mail. At the same time the dashboard picks up the new record, so the KPI tiles and charts are always current.

4.5 User Interface and Navigation Design

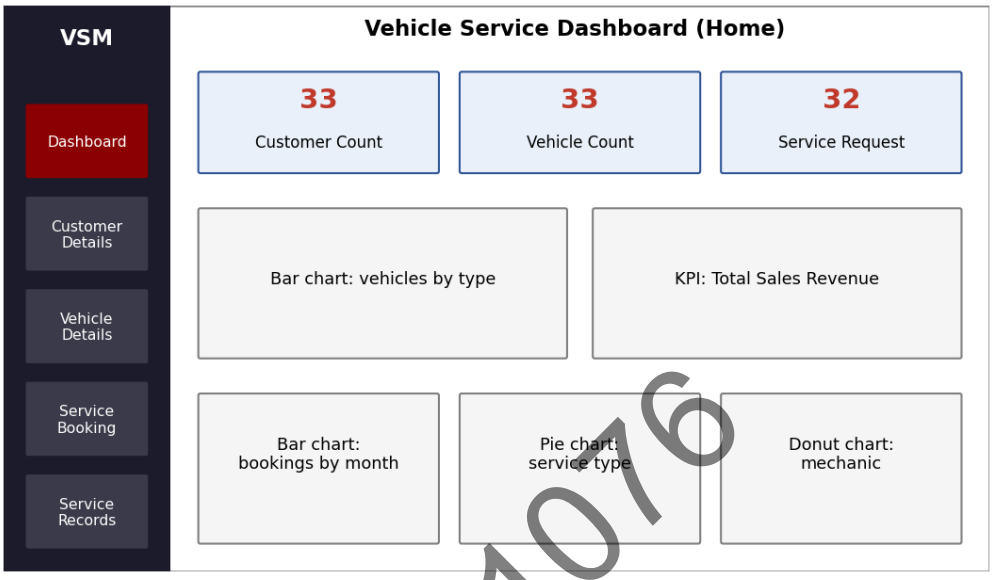


Figure 4.4: Navigation Structure and Wireframe of the Vehicle Service Dashboard Page

The application has a dark left-hand menu with five entries: the Vehicle Service Dashboard (Home), customer details, Vehicle Details, Service Booking and Service Records. The last four entries open the form for data entry and the report that lists the saved records. The dashboard is laid out with the three count tiles at the top, the vehicle-type chart and the total sales revenue tile below them, and the booking, service-type and mechanic charts at the bottom, as shown in Figure 4.4.

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4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Mechanic
Form: customer details	A E V D	V
Form: Vehicle Details	A E V D	V
Form: Service Booking	A E V D	A E V
Form: Service Records	A E V D	A E V
Reports (all four)	V	V
Page: Vehicle Service Dashboard	V	V
Users and permissions	Manage	—

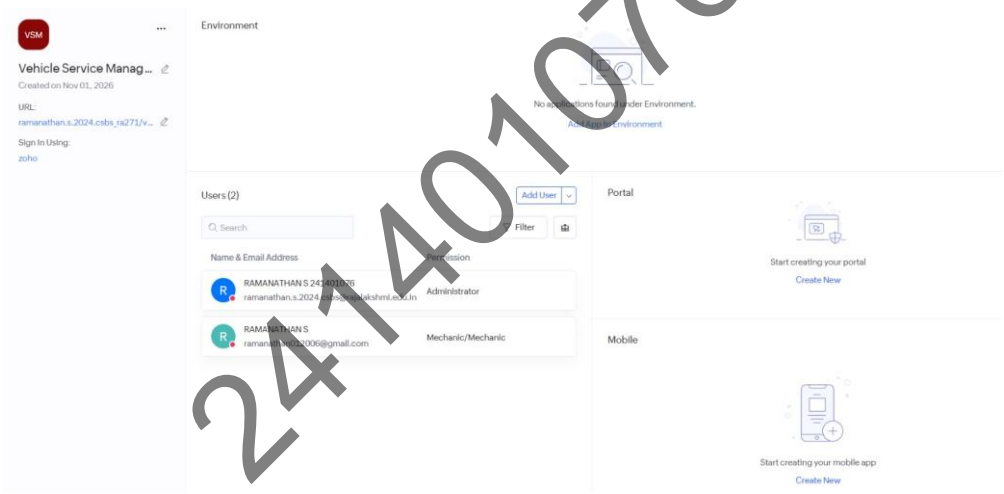
A = add, E = edit, V = view, D = delete. The matrix shows the designed access; the two users and their permissions are listed on the application summary page (Figure 5.1).

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CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in the Zoho Creator workspace of the account ramanathan.s.2024.csbs_ra271 under the name “Vehicle Service Management System” on 01 November 2026 (Figure 5.1). Its sign-in method is Zoho, and the application summary lists two users. The Environment, Portal and Mobile panels of the summary are empty, because no environment, customer portal or mobile app was configured. The Zoho Creator Learning Centre was used as a guide for the builder [7]. The dark left-hand menu supplied by the platform was kept and organised into five sections: the Vehicle Service Dashboard (Home), customer details, Vehicle Details, Service Booking and Service Records. The Zoho Creator trial edition was used.



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Figure 5.1: Application Summary, URL and Users

5.2 Forms and Data Capture

The application has four forms, whose fields are listed in Table 4.2. The customer details form (Figure 5.2) captures the Customer name, a Phone number with a country-code selector that defaults to +91, the Email and the Address. The Vehicle Details form selects the owner from the customer lookup and then takes the Vehicle Number, Brand, Model, Vehicle Type, Fuel Type and Manufacturing Year; Brand, Vehicle Type and Fuel Type are drop-down fields so that every user picks from the same values.

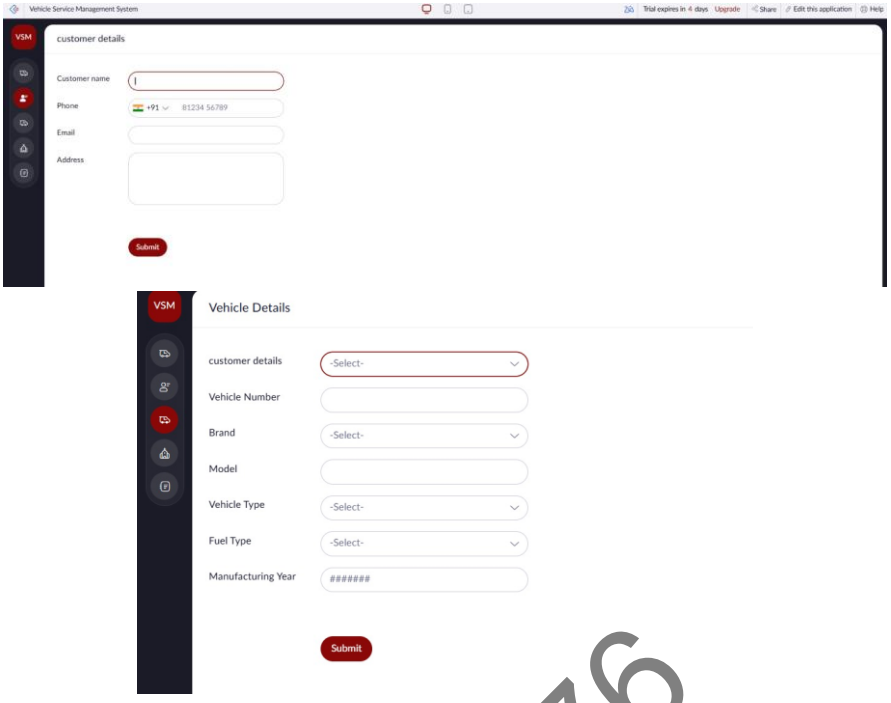


Figure 5.2: customer details and Vehicle Details Forms

The Service Booking form (Figure 5.3) links a booking to a customer and a vehicle through two lookup drop-downs and then captures the Service Type, the Booking Date in the dd-MMM-yyyy format, the Preferred Time slot, the Problem Description, the Estimated Cost and the Booking Status. The Service Records form links the work to a booking and records the Assigned Mechanic, the Service Date, the Work Description, the Parts Cost, the Labour Cost, the Total Amount, the Service Status and the Payment Status. In every record visible in Figure 5.5 the Total Amount equals the Parts Cost plus the Labour Cost. Each form has a Submit button. Thirty-three customers, 33 vehicles, 32 bookings and 32 service records were entered in these forms.

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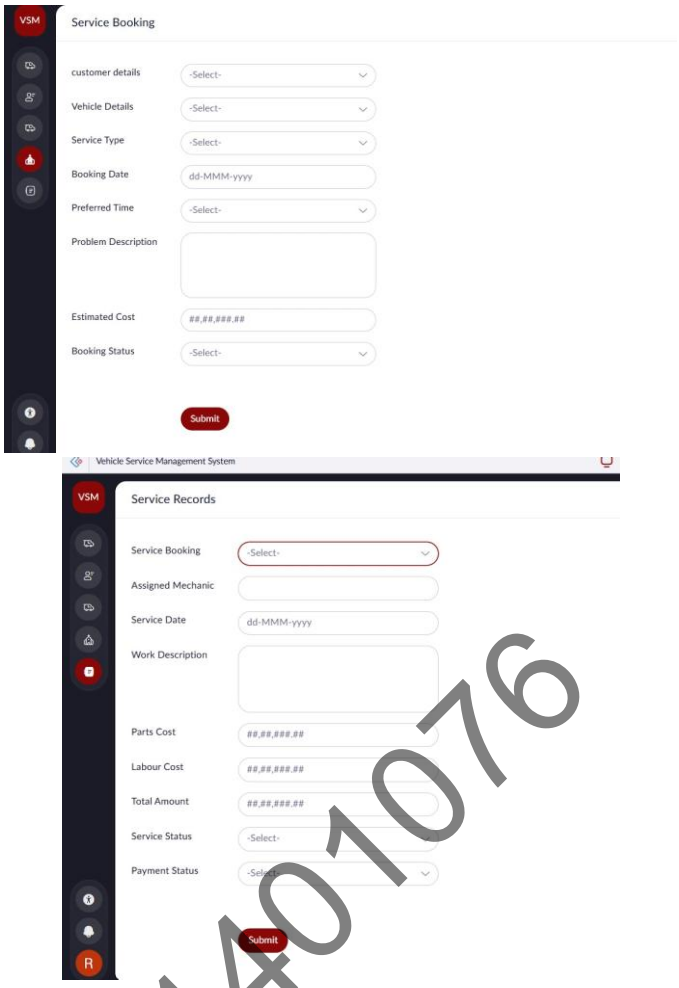


Figure 5.3: Service Booking and Service Records Forms

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5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Columns Shown	Records
All Customer Details	List	customer details	Customer name, Phone, Email, Address	33 of 33
All Vehicle Details	List	Vehicle Details	customer details, Vehicle Number, Brand, Model, Vehicle Type, Fuel Type, Manufacturing Year	33 of 33
Service Booking Report	List	Service Booking	customer details, Vehicle Details, Service Type, Booking Date, Preferred Time, Problem Description, Estimated Cost	32 of 32
All Service Records	List	Service Records	Service Booking, Assigned Mechanic, Service Date, Work Description, Parts Cost, Labour Cost, Total Amount	32 of 32

Four list reports were built, one on each form. Each report shows the records in a table with sortable columns, a search button, a plus button that opens the form and a menu

button, and the footer reports how many records are shown, for example “Showing 33 of 33”. All Customer Details and All Vehicle Details are shown in Figure 5.4, and the Service Booking Report and All Service Records in Figure 5.5. Kanban, calendar and pivot reports were not needed for this scope.



Figure 5.4: All Customer Details and All Vehicle Details Reports

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Figure 5.5: Service Booking Report and All Service Records Reports

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5.4 Pages and Dashboards

The Vehicle Service Dashboard (link name Vehicle_Service_Dashboard) is a page built with the UI Builder and is the application's Home. It is intended for the Administrator and shows three count tiles for Customer Count (33), Vehicle Count (33) and Service Request (32), a bar chart of vehicles by type (20 cars, 11 bikes and 2 scooters), a Total Sales Revenue tile of ₹519,700.00, a bar chart of bookings by month (30 in Nov 2026 and 2 in Dec 2026), a pie chart of bookings by service type and a donut chart of service records by mechanic (Figure 5.6). The page reads directly from the forms, so it needs no refresh script, and the three count tiles match the record counts of the reports in Figures 5.4 and 5.5.

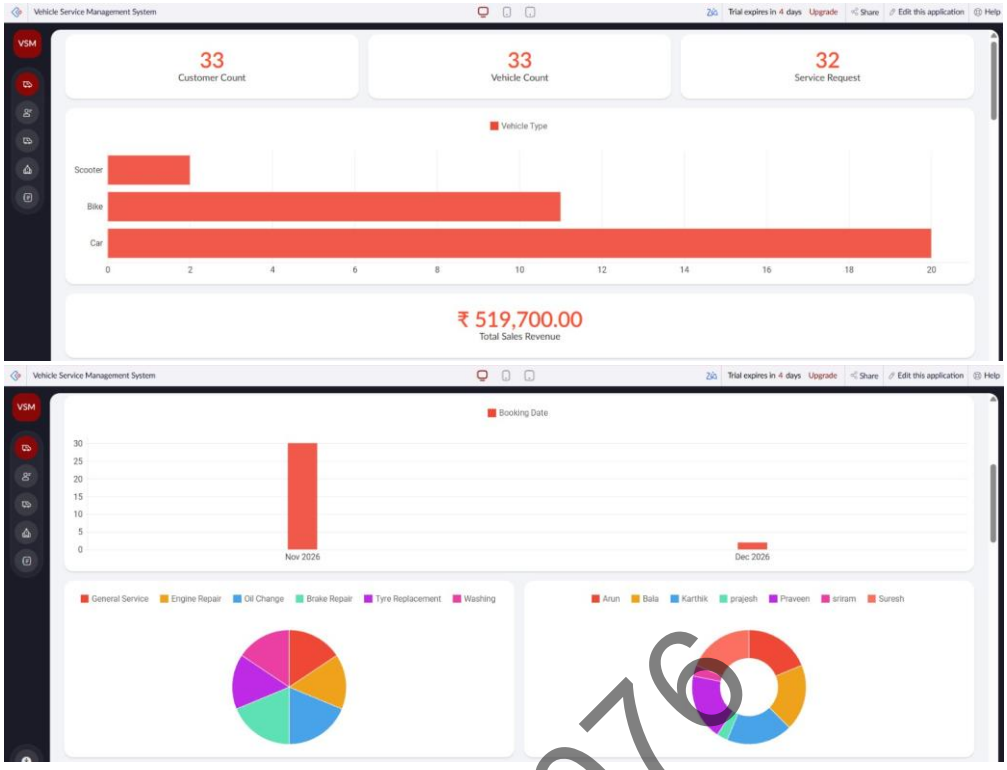


Figure 5.6: Vehicle Service Dashboard

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Workflow: booking confirmation → Successful form submission	Look up the service package, work out the delivery date, set the status to Booked and e-mail the owner	fetch with criteria, if / else if, string contains(), toLowerCase(), addDay(), date toString(), sendmail task, zoho.adminuserid

Code 5.1: Booking confirmation script (DS-01)

```
// 1. Get package details from the Service_Packages lookup
packageName = "";
packageCost = "";
days = 0;
pkgRec = Service_Packages[ID == input.Service_Packages];
if(pkgRec.count() > 0)
{
    packageName = pkgRec.Package_Name;
    packageCost = pkgRec.Service_Cost;

    // Days needed, worked out from the package name
    pn = packageName.toString().toLowerCase();
    if(pn.contains("basic"))
    {
        days = 1;
    }
```

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```
    else if(pn.contains("standard"))
    {
        days = 2;
    }
    else if(pn.contains("premium"))
    {
        days = 3;
    }
    else if(pn.contains("full"))
    {
        days = 5;
    }
}

// 2. Calculate delivery date if it was left blank
serviceDate = input.Service_Date;
deliveryDate = input.Delivery_Date;
bookRec = Vehicle_Service_Booking[ID == input.ID];
if(deliveryDate == null && serviceDate != null && days > 0)
{
    deliveryDate = serviceDate.addDay(days);
    bookRec.Delivery_Date = deliveryDate;
}

// 3. Set status to Booked if it was left blank
if(input.Service_Status == null || input.Service_Status == "")
{
    bookRec.Service_Status = "Booked";
}

// 4. Send the confirmation email
if(input.Email_Address != null && input.Email_Address != "")
{
    serviceText = "";
    deliveryText = "";
    if(serviceDate != null)
    {
        serviceText = serviceDate.toString("dd-MMM-yyyy");
    }
    if(deliveryDate != null)
    {
        deliveryText = deliveryDate.toString("dd-MMM-yyyy");
    }

    sendmail
    [
        from: zoho.adminuserid
        to: input.Email_Address
        subject: "Service Booking Confirmed, " + input.Owner_Name +
"!"
        message: "Dear " + input.Owner_Name + ",<br><br>"
               + "Thank you for booking with our service center. Your
booking details are below:<br><br>"
               + "<b>Vehicle Number:</b> " + input.Vehicle_Number +
"<br>"
               + "<b>Service Package:</b> " + packageName + "<br>"
               + "<b>Service Cost:</b> Rs. " + packageCost + "<br>"
               + "<b>Service Date:</b> " + serviceText + "<br>"
               + "<b>Expected Delivery:</b> " + deliveryText +
"<br><br>"
               + "We look forward to servicing your vehicle.<br><br>"
               + "Regards,<br>Vehicle Service Management Team"
```

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```
    ]
}
```

DS-01 runs when a booking is submitted. It has four parts. Part 1 fetches the service package record chosen in the Service_Packages lookup with a criteria on its ID and reads the package name and cost; the number of days needed is then worked out from the package name: basic 1, standard 2, premium 3 and full 5. Part 2 calculates the expected delivery date by adding those days to the service date, but only when the delivery date was left blank. Part 3 sets the status to “Booked” when it was left blank. Part 4 sends the confirmation e-mail, but only when an e-mail address is present; the dates are formatted as dd-MMM-yyyy, the e-mail is sent from the administrator’s address (zoho.adminuserid), and the message lists the vehicle number, service package, service cost, service date and expected delivery. The script is attached to the form workflow described in Section 5.6.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Booking confirmation mail	Form Workflow	Created → Successful form submission	None (execute without condition)	Run the Deluge script: delivery date, default status and confirmation e-mail	DS-01

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One form workflow was created on the Service Booking form with the record event Created and the form event Successful form submission. Its action is the Deluge script of Code 5.1, which fills in the delivery date and the status and sends the booking confirmation e-mail to the vehicle owner through the Send Mail task. Schedule, approval, Blueprint, batch and payment workflows were not required for this scope, which matches the counts in the Project at a Glance table.

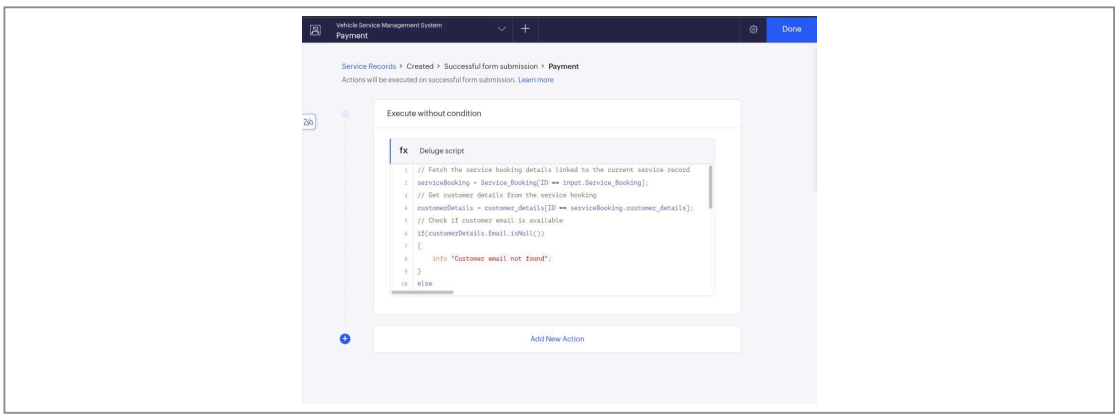


Figure 5.7: Booking Confirmation Mail Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Creator mail service	Send Mail (sendmail task) in the booking workflow, sent from zoho.adminuserid	Send Mail action inside the Deluge script	Managed by the platform	Booking confirmation e-mail to the vehicle owner

The only integration is Zoho Creator's own mail service. No external API call, REST endpoint, webhook or connection was configured, so no API key, client secret or token is stored in the application.

5.8 User Management and Role-Based Access

Users were configured as designed in Table 4.3 and are listed on the application summary page (Figure 5.1). The Administrator is the application owner, RAMANATHAN S 241401076 (ramanathan.s.2024.csbs@rajalakshmi.edu.in). One further user, RAMANATHAN S (ramanathan012006@gmail.com), was added with the Mechanic permission, so the application has two users. Both users sign in with their Zoho accounts. The Administrator manages users from the summary page, and the Mechanic works with the booking and service-record data. No separate role hierarchy or data sharing rule was needed for a two-user application.

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5.9 AI and Analytics Features

Analytics in this project are the KPI tiles and charts of the Vehicle Service Dashboard: count tiles for customers, vehicles and service requests, a Total Sales Revenue tile, a bar chart of vehicles by type, a bar chart of bookings by month, a pie chart of bookings by service type and a donut chart of service records by mechanic (Figure 5.6). They are built with the page builder and calculated from the live records of the forms. No Zia or other AI feature, such as prediction, OCR or sentiment analysis, was used. The dashboard figures were checked against the reports: 33 customers, 33 vehicles and 32 service requests agree with the record counts of the reports in Figures 5.4 and 5.5, and the vehicle-type values (20 + 11 + 2) add up to 33.

5.10 Pro-Code Extension

No Zoho Catalyst service, JavaScript widget, custom client script or external API call was added, because the requirements are fully met by forms, reports, a page and one

workflow. If the system grows, Catalyst could host a microservice for predictive service reminders, and widgets could provide a customised job board (Section 9.2).

5.11 Mobile and Portal Access

The application was built and tested in the web browser. Zoho Creator publishes the same application to its Android and iOS mobile app automatically, and the builder toolbar offers desktop, tablet and phone previews, but no mobile-specific layout, offline setting or customer portal was configured; the Portal and Mobile panels in Figure 5.1 are empty.

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CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at four levels. Unit testing checked each of the four forms on its own. Integration testing checked that the lookup fields link the forms, that the reports show the saved records and that the dashboard figures agree with the reports and that the booking workflow sends the confirmation e-mail. Role-based testing used the two accounts (Administrator and Mechanic) to confirm the users and permissions. Acceptance testing was done by the same two users in their assigned roles (Section 6.5).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add a customer	Open customer details; enter name, phone, e-mail, address; Submit	Record saved and listed	Customer listed in All Customer Details (Showing 33 of 33)	Pass
TC-02	FR-02	Register a vehicle for a customer	Open Vehicle Details; select the customer; enter number, brand, model, type, fuel, year; Submit	Vehicle saved against the customer	Vehicle listed with its customer in All Vehicle Details (Showing 33 of 33)	Pass
TC-03	FR-03	Book a service	Open Service Booking; select customer, vehicle, service type, date, time; enter problem and cost; Submit	Booking saved and listed	Booking listed in the Service Booking Report (Showing 32 of 32)	Pass
TC-04	FR-04	Record the service work	Open Service Records; select the booking; enter mechanic, date, work, parts and labour cost; Submit	Record saved; total equals parts plus labour	Records listed in All Service Records (Showing 32 of 32); total correct in all visible rows	Pass
TC-05	FR-05	Confirmation e-mail on booking	Submit a booking with the owner's e-mail address and open the inbox	E-mail with vehicle number, package, cost and dates	[To be completed with the workflow / e-mail screenshot]	Pending
TC-06	FR-06	Reports show all records	Open the four reports and read the footer counts	All saved records listed	33, 33, 32 and 32 records shown	Pass
TC-07	FR-07	Dashboard KPI tiles	Open Home and read the tiles	Counts and revenue shown	Customer Count 33, Vehicle Count 33, Service Request 32, Total Sales Revenue ₹519,700.00	Pass

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TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-08	FR-07	Dashboard charts	Open Home and read the charts	Charts rendered from live data	Vehicle type: Car 20, Bike 11, Scooter 2; bookings: Nov 2026 = 30, Dec 2026 = 2; pie and donut rendered	Pass
TC-09	FR-08	Users and permissions	Open the application summary and read Users (2)	Administrator and Mechanic listed	Both users listed with the expected permissions	Pass

6.3 Debugging Deluge Scripts

Debugging relied on the platform's own tools: the syntax check of the Deluge editor when the script is saved, optional info statements for inspecting values such as packageName, days and deliveryDate, and the saved booking record, which shows whether the delivery date and the status were filled in. For the data, the record count in the footer of each report was compared with the matching dashboard tile. The issues noticed while testing are recorded in Table 6.2.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG-01	A test vehicle was saved with the Manufacturing Year 2040 and another with 2000	The Manufacturing Year number field has no range validation	Add a validation so the year cannot be later than the current year	Open
BUG-02	Vehicle numbers appear in different formats (for example "tn 13 jj 1234" and "TN39EF2030")	Vehicle Number is a free-text field without a format rule	Add a pattern validation and convert entries to upper case	Open
BUG-03	Mechanic names are typed in different styles in Service Records, so the same person could be counted twice in the donut chart	Assigned Mechanic is a single-line text field, not a lookup	Replace the field with a lookup or drop-down of mechanics	Open

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6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Ramanathan S (Administrator)	Administrator	Forms, reports and dashboard are clear; the dashboard counts agree with the reports	None required
Ramanathan S (Mechanic account)	Mechanic	The Service Records form is short and records parts, labour and total in one place	None required

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CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and published directly from the account's default workspace and is live at https://creatorapp.zoho.in/ramanathan.s.2024.csbs_ra271/vehicle-service-management-system/ since 01 November 2026. The Environment panel of the application summary shows "No applications found under Environment", so separate Stage and Production environments were not configured for this trial project (Figure 5.1). Changes were tested first in the live builder with test records and then with the two user accounts. As a rollback plan, the application can be exported as a backup before any structural change.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in the development workspace	✓
2	Sample data kept for the demonstration and identified as test data	✓
3	Users and permissions configured (Administrator and Mechanic)	✓
4	Booking confirmation workflow created	✓
5	Application published and URL verified	✓
6	Trial expiry noted (the trial banner shows 4 days remaining)	✓

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7.2 Performance Optimisation

Because the data set is small, performance work was limited to good design practice, mainly reducing the number of entries a user has to make. Table 7.2 summarises the effect.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Customer and vehicle chosen through lookup fields instead of being typed again	Service Booking form	10 fields to type (4 customer + 6 vehicle)	2 lookup selections	8 fewer entries per booking
Delivery date and status filled in by the script	Booking workflow	Delivery date and status entered by hand	Filled in automatically when blank	2 manual entries removed
Dashboard built directly on the form data	Vehicle Service Dashboard	Counts and revenue totalled by hand from registers	Live tiles and charts	No manual counting

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Two configured users with separate permissions (Administrator, Mechanic)	Application users	Unauthorised access to or changes of records
Zoho account sign-in with OTP re-authentication where requested	Application login	Account takeover
Drop-down choices for Brand, Vehicle Type, Fuel Type, Service Type and Preferred Time	Forms	Invalid or inconsistent values
E-mail sent from the administrator's address (zoho.adminuserid) through the platform mail service	Booking workflow	Spoofed or untraceable sender

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No hard-coded password, API key or client secret is stored in the application. Customer phone numbers, e-mail addresses and home areas are personal data, so access is limited to signed-in users. Encrypted fields were not needed, because no confidential identifiers such as card or identity numbers are stored.

7.4 Governance and Administration

The administrator manages the two users from the application summary page and can add or remove users and change their permissions at any time. Forms, reports and the dashboard carry meaningful names, and the dashboard page has the link name Vehicle_Service_Dashboard. Record changes are traceable through the audit trail, a copy of the application can be exported as a backup, and the number of users is governed by the licence of the plan; the trial edition was used and expires after 15 days from the start. The application owner is responsible for maintenance and for reviewing users.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The output follows the data of a service job through the application. Figure 8.1 shows the All Customer Details report with customers such as Pooja M, Sridhar R and Varsha K and their phone numbers, e-mail addresses and Chennai areas (Showing 33 of 33). Figure 8.2 shows All Vehicle Details, where each vehicle is linked to its owner, for example the Tata Tiago TN39EF2030 of Pooja M and the Ather 450S TN37EF2028 of Varsha K (Showing 33 of 33). Figure 8.3 shows the Service Booking Report with the service type, booking date, preferred time, problem description and estimated cost of each booking, for example a Washing booking on 02-Dec-2026 for ₹650.00 (Showing 32 of 32). Figure 8.4 shows All Service Records with the assigned mechanic, work description, parts cost, labour cost and total amount, for example record 1030 by Suresh with parts ₹1,000.00, labour ₹700.00 and a total of ₹1,700.00 (Showing 32 of 32).

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Figure 8.1: Output – All Customer Details Report (33 Customers)

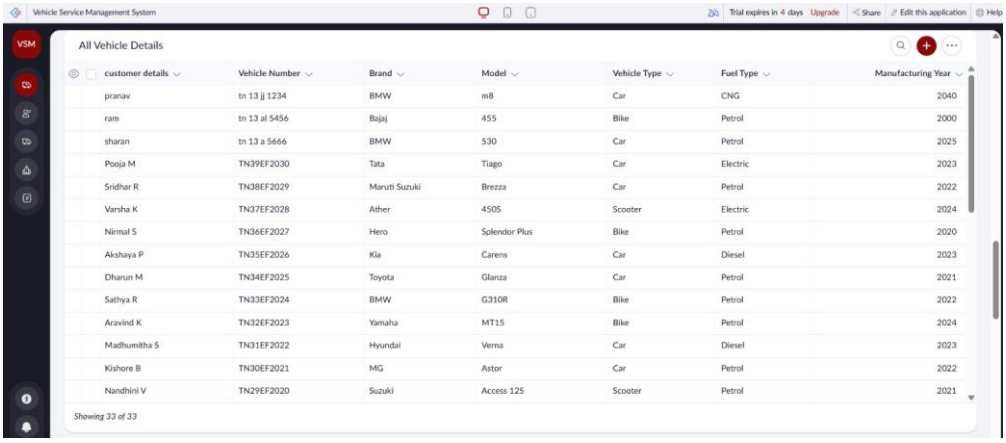


Figure 8.2: Output – All Vehicle Details Report (33 Vehicles)

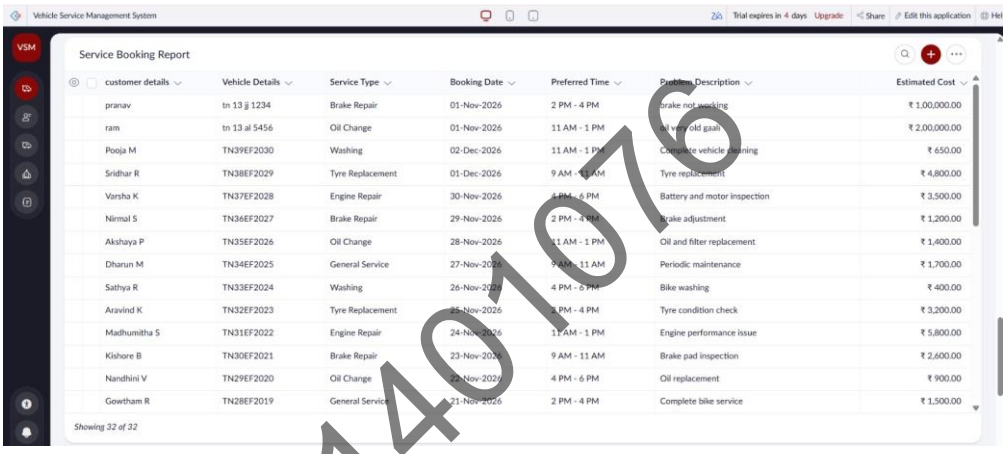


Figure 8.3: Output – Service Booking Report (32 Bookings)

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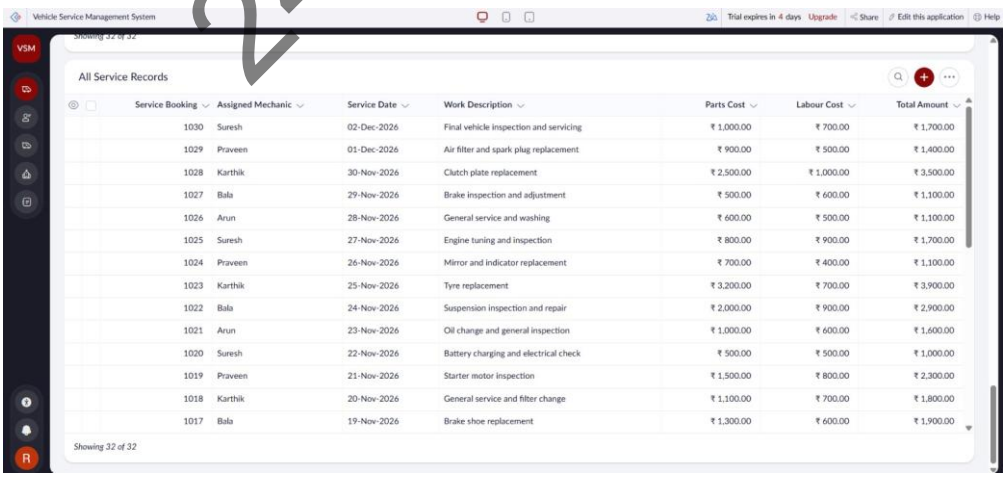


Figure 8.4: Output – All Service Records Report (32 Records)

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To design a linked data model and forms for the service records	Section 5.2, Table 4.2	Four forms linked by lookups; 33 customers, 33 vehicles, 32 bookings and 32 service records stored
To automate booking confirmation	Sections 5.5 and 5.6	Booking workflow with the DS-01 script (Code 5.1)
To provide reports and a dashboard	Sections 5.3 and 5.4	Four list reports and the Vehicle Service Dashboard (Figures 5.4 to 5.6)
To send e-mail through the mail service and restrict access by user	Sections 5.7 and 5.8	Send Mail action; Administrator and Mechanic users (Figure 5.1)
To publish the application with governance controls	Section 5.1 and Chapter 7	Published at the live URL on 01 November 2026 with two users

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one booking	Writing customer and vehicle details in a register and on a job card	One booking form using lookups; confirmation e-mail sent automatically	Fewer entries and an immediate confirmation
Manual steps	Write register, write job card, total costs, phone the owner, count totals	Enter the booking and the service record; e-mail, delivery date and totals are automatic	Several manual steps removed
Report generation	Manual counting of vehicles, jobs and revenue from registers	Ready list reports and a live dashboard	Real-time, no manual effort
Data accessibility	Paper registers kept at the centre	Cloud application at one URL with signed-in users	Anywhere, with controlled access

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8.4 Limitations of the Low-Code Approach

Several limits of the low-code approach were met. The trial edition is time-limited (the banner in the screenshots shows 4 days remaining), which forces the project to be finished and demonstrated within that period. The UI of forms and pages can be customised only within the options of the builder, so the charts follow the platform's styling. Deluge has statement and API-call limits, so scripts must stay short and use one

fetch where possible. The application depends on Zoho's cloud, which means a degree of vendor lock-in. Free-text fields such as Vehicle Number and Assigned Mechanic allow inconsistent entries (Table 6.2) unless validation or lookups are added. These limits were handled by keeping the design simple, using built-in components first and writing code only where it was unavoidable.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the no-code → low-code → pro-code spectrum, most of the Vehicle Service Management System is no-code configuration: the four forms, the lookup and drop-down fields, the four list reports, the dashboard page and the user setup. Low-code appears in the Deluge script DS-01 that calculates the delivery date, sets the default status and sends the confirmation e-mail. No pro-code was used. A move to pro-code, for example with Zoho Catalyst, would be justified if the system needed predictive maintenance, complex multi-system integrations or a fully custom user interface.

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CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Vehicle Service Management System replaces a register-based service process with a Zoho Creator application made of four linked forms, four list reports, one dashboard page, one workflow with a Deluge script and two users. Thirty-three customers, 33 vehicles, 32 bookings and 32 service records are stored, the dashboard shows the live counts and a total sales revenue of ₹519,700.00, and the application was created and published on 01 November 2026 at its live URL.

All five objectives of Section 1.3 were met, as shown in Table 8.1. The project also showed that a working business application can be built quickly on a low-code platform, with code used only where it adds value.

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9.2 Future Enhancements

- Add validation rules (Manufacturing Year not later than the current year, Vehicle Number format) and replace the Assigned Mechanic text field with a lookup, to remove the issues of Table 6.2.
- Add a scheduled workflow that e-mails or messages owners about the next service due date and about pending payments, and an approval process for estimates above a limit.
- Add a customer portal for online booking and status tracking, payment collection, and Zia-based predictive maintenance together with a mobile layout for mechanics in the workshop.

9.3 Learning Outcomes and Career Relevance

Through this project I learned how to model linked data in Zoho Creator forms with lookup fields, build list reports, design a dashboard page with KPI tiles and charts, create a form workflow with a Deluge script that sends e-mail, and set up users with permissions. These skills correspond to the roles of citizen developer, low-code developer and automation engineer, and the experience of building and testing a live application is directly useful for a solutions-consultant role in a business-systems team.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Vehicle Service Management System; menu: Home dashboard, customer details, Vehicle Details, Service Booking, Service Records	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Single Line, Phone, Email, Multi Line, Drop Down, Lookup, Number, Date and Currency across four forms	4.3, 5.2	Figure 5.2, 5.3
II	Lookup field (relationship between forms)	Y	customer details, Vehicle Details and Service Booking lookups	4.2, 5.2	Figure 5.2, 5.3
II	Subform (parent–child data)	N	Not required; forms are linked by lookups	4.2	—
II	Field validations / layout rules	N	Drop-down choices only; no layout rules or range validation (Table 6.2)	5.2	Table 6.2
II	List report with filter, grouping, custom action	N	Four plain list reports (no custom action configured)	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	—
II	Pivot table / chart report	N	Charts are on the dashboard page, not a pivot report	5.3	—
II	Page / dashboard built with UI Builder	Y	Vehicle Service Dashboard (Home)	5.4	Figure 5.6
II	Deluge – form events (validate / success / field actions)	Y	Booking workflow (Successful form submission)	5.5	Code 5.1
II	Deluge – functions, fetch, loops, collections	N	Fetch and conditions used in the workflow script; no standalone function, loop or collection	5.5	Code 5.1
II	Email / in-app notifications	Y	Booking workflow → sendmail	5.5, 5.6	Code 5.1
II	Form workflow	Y	Booking confirmation mail workflow	5.6	Table 5.3, Figure 5.7
II	Scheduled workflow	N	Not required	5.6	—
II	Approval process	N	Not required	5.6	—
II	Blueprint	N	Not required	5.6	—
II	Batch / report workflow	N	Not required	5.6	—

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Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Testing & debugging evidence	Y	Table 6.1 and Table 6.2	6	Table 6.1
III	Integration with a Zoho service	Y	Zoho Creator mail service	5.7	Code 5.1
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	Two users: Administrator and Mechanic	5.8	Figure 5.1
III	Record-level access criteria / customer portal	N	Not configured	5.8	—
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard count tiles, bar, pie and donut charts	5.9	Figure 5.6
IV	Performance optimisation applied & measured	Y	Lookups, automatic delivery date and status (entries reduced)	7.2	Table 7.2
IV	Deployment across environments	N	Single workspace; no Stage or Production environment	7.1	Figure 5.1
IV	Security controls (encryption, OAuth, permissions)	Y	User permissions, Zoho sign-in, drop-down values	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	User administration and audit trail	7.4	Figure 5.1
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	N	Not configured	5.11	—

Total features implemented: 15 / 32

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APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

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APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Booking Confirmation – Workflow Script

```
// Script ID : DS-01
// Location  : Form workflow on Service Booking → Successful form
submission
// Purpose   : Fill the delivery date and status, then e-mail the
booking confirmation
// 1. Get package details from the Service_Packages lookup
packageName = "";
packageCost = "";
days = 0;
pkgRec = Service_Packages[ID == input.Service_Packages];
if(pkgRec.count() > 0)
{
    packageName = pkgRec.Package_Name;
    packageCost = pkgRec.Service_Cost;

    // Days needed, worked out from the package name
    pn = packageName.toString().toLowerCase();
    if(pn.contains("basic"))
    {
        days = 1;
    }
    else if(pn.contains("standard"))
    {
        days = 2;
    }
    else if(pn.contains("premium"))
    {
        days = 3;
    }
    else if(pn.contains("full"))
    {
        days = 5;
    }
}

// 2. Calculate delivery date if it was left blank
serviceDate = input.Service_Date;
deliveryDate = input.Delivery_Date;
bookRec = Vehicle_Service_Booking[ID == input.ID];
if(deliveryDate == null && serviceDate != null && days > 0)
{
    deliveryDate = serviceDate.addDay(days);
    bookRec.Delivery_Date = deliveryDate;
}

// 3. Set status to Booked if it was left blank
if(input.Service_Status == null || input.Service_Status == "")
{
    bookRec.Service_Status = "Booked";
}

// 4. Send the confirmation email
if(input.Email_Address != null && input.Email_Address != "")
{
    serviceText = "";
    deliveryText = "";
    if(serviceDate != null)
    {
```

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```
        serviceText = serviceDate.toString("dd-MMM-yyyy");
    }
    if(deliveryDate != null)
    {
        deliveryText = deliveryDate.toString("dd-MMM-yyyy");
    }

    sendmail
    [
        from: zoho.adminuserid
        to: input.Email_Address
        subject: "Service Booking Confirmed, " + input.Owner_Name +
"!"
        message: "Dear " + input.Owner_Name + ",<br><br>"
               + "Thank you for booking with our service center. Your
booking details are below:<br><br>"
               + "<b>Vehicle Number:</b> " + input.Vehicle_Number +
"<br>"
               + "<b>Service Package:</b> " + packageName + "<br>"
               + "<b>Service Cost:</b> Rs. " + packageCost + "<br>"
               + "<b>Service Date:</b> " + serviceText + "<br>"
               + "<b>Expected Delivery:</b> " + deliveryText +
"<br><br>"
               + "We look forward to servicing your vehicle.<br><br>"
               + "Regards,<br>Vehicle Service Management Team"
    ]
}
```

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APPENDIX D – USER MANUAL

D.1 Accessing the Application

Web: open https://creatorapp.zoho.in/ramanathan.s.2024.csbs_ra271/vehicle-service-management-system/ in Chrome, Edge or Firefox and sign in with the Zoho account that the Administrator added (Mechanic: ramanathan012006@gmail.com). If Zoho asks for an OTP, enter the code sent to your e-mail. Mobile: install the Zoho Creator app, sign in with the same account and open the application from the list.

The Administrator signs in with ramanathan.s.2024.csbs@rajalakshmi.edu.in and, in addition to the steps below, manages users from the application summary page (Add User → enter e-mail → choose a permission) and reviews the booking workflow under Workflow.

D.2 Instructions for Administrator

- Open customer details from the left-hand menu, enter the Customer name, Phone, Email and Address and click Submit.
- Open Vehicle Details, select the customer, enter the Vehicle Number and Model, choose the Brand, Vehicle Type and Fuel Type, enter the Manufacturing Year and click Submit.
- Open Service Booking, select the customer and the vehicle, choose the Service Type and Preferred Time, enter the Booking Date, the Problem Description and the Estimated Cost, and click Submit; the confirmation e-mail is sent automatically.
- Open Home to see the Vehicle Service Dashboard with the count tiles, the revenue tile and the four charts.
- Use the four reports (All Customer Details, All Vehicle Details, Service Booking Report, All Service Records) to review and search the saved records.

D.3 Instructions for Mechanic

- Open Service Booking Report to see the bookings, their service type, date, preferred time and problem description.
- Open Service Records, select the booking, enter the Assigned Mechanic, Service Date and Work Description, and enter the Parts Cost, Labour Cost and Total Amount.

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- Set the Service Status and Payment Status and click Submit; wait for the record to appear in All Service Records.
- Open All Vehicle Details to check the brand, model, vehicle type and fuel type of the vehicle before the work.
- Open Home to check the dashboard counts and the number of jobs per mechanic.

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APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	RAMANATHAN S	Register No.	241401076
Project Title	Vehicle Service Management System	Date	03 November 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

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Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

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APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	RAMANATHAN S	Register No.	241401076
Department / Section	B.Tech CSBS	Date of Viva	03 November 2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

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Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____